UNIVERSITY OF
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To Whom it May Concern,

My name is Dr. Muriel C. Maurer, and I am a professor of biochemistry at the University of Louisville. I received my B.A. from Goucher College in 1987. In 1993, I earned my Ph.D. from the University of Virginia, before completing 4 years of postdoctoral study and a year as a research associate at Cornell University. Generally speaking, my research centers on protein chemistry, especially in terms of structure-function relationships. I also specifically investigate the enzymes related to blood coagulation and the role they play in various conditions, including stroke, heart disease, and cancer. I am writing to you today in order to provide a discussion of the qualifications and research of Dr. Akul Yugesh Mehta, an outstanding and prolific researcher in the field of biotechnologies.

I am most familiar with Dr. Mehta's research centered on developing novel, synthetic replacements for the drug heparin. This drug is commonly used during surgery due to its properties as an anticoagulant in order to prevent the development of blood clots (process called coagulation) and associated heart attacks, however it has also been associated with significant bleeding complications. In addition, heparin is a sulfated polymeric carbohydrate mixture usually derived from animal sources, hence a synthetic alternative offers more consistency in its production and therefore greater safety. Heparin is known to target exosites on coagulation enzymes such as thrombin, in order to inhibit them. Dr. Mehta therefore developed a series of assays for characterizing inhibition of such exosites with sulfated small molecules in order to discover a suitable synthetic replacement. He used these assays to evaluate the inhibition of coagulation enzymes for the various molecules which he synthesized, as well as their affinity for binding to various enzymes. Through these efforts, Dr. Mehta discovered sulfated beta-O-4 lignin, or SbO4L, a highly unique anticoagulant and antithrombotic. He found that this molecule is able to inhibit thrombin using allosteric binding at a site associated with platelet receptor GPIbalpha and heparin, thereby demonstrating a unique dual anticoagulation and antiplatelet activity. He additionally identified a group known as sulfated benzofuran small molecules, which also possess excellent capacities as anticoagulants, as well as other molecules such as sulfated coumarin-type molecules and sulfated pentagalloylglucoside, which target another enzyme of coagulation known as Factor XIa. Through these efforts, Dr. Mehta has presented both highly valuable biomolecules with important applications and particularly useful assays for characterizing similar molecules.

Many researchers from around the world have benefited substantially from these findings. In fact, I count myself among these researchers, as I have cited Dr. Mehta's research on two separate occasions. Dr. Mehta identified small molecule inhibitors of allosteric exosite II on thrombin which show potent anticoagulation and reduced bleeding, a fact which led me to further investigate and evaluate peptides that target these sites and to understand the structural mechanism between the sites. I cited Dr. Mehta in order to support this work and successfully identified several novel mechanisms at play in these sites.

Blood clots represent a serious surgical and post-surgical complication with the capacity to cause deadly consequences, including heart attacks, strokes, and pulmonary embolisms. Millions of patients receive heparin every single year to prevent this problem, but as noted above, heparin is associated with severe side effects that impact many of these patients. Anticoagulants such as heparin are also used in preventing these problems outside of the surgical setting, providing life-

saving support for patients at risk of blood clots and the associated deadly consequences. Through his outstanding work, Dr. Mehta has provided an excellent pathway towards improving these therapies and removing their side effects, thereby making new molecules which are substantially safer and more effective for use.

Notably, this research represents only a portion of Dr. Mehta's total body of work. Across his career, he has produced and published dozens of articles in important journals for his field, including *Science Advances, Journal of Thrombosis and Haemostasis,* and *Cellular and Molecular Life Sciences*. These journals receive far more submissions than they are capable of publishing and subject all manuscripts to rigorous review procedures in order to confirm the research is sound. This record therefore serves as a clear acknowledgement of the value that Dr. Mehta's research holds for the wider field.

To conclude, Dr. Mehta is a highly talented researcher who has successfully produced novel approaches for evaluating and characterizing a class of important molecules and discovered key examples within this group with major applications as anticoagulants. Dr. Mehta's expertise in the field is further confirmed through his excellent publication record. Collectively, these factors show that Dr. Mehta is a leading researcher for the field of biotechnologies, and that he has earned your support in his endeavors. Please feel free to contact me if you have further questions regarding Dr. Mehta's research.

With kind regards,

Muriel C Maurer

Muriel C. Maurer
Professor of Chemistry